

Chapter 10

Resistance

RESISTANCE:

Definition	Resistance is a measure of the opposition to the flow of electric current.
Reason	Collision of free electrons with the atoms of conductors
Symbol	R
S.I. Unit	Ohm (Ω)
Circuit symbol	
Factors affecting	Nature, dimensions and temperature of conductor

OHM'S LAW:

In 1826, German physicist GEORGE SIMON OHM (16 March 1789 – 6 July 1854) given a quantitative relation between the potential difference across the ends of a conductor and electric current flowing through it.

Statement

The electric current passing through a conductor is directly proportional to the potential difference between the ends of conductor, if physical conditions of conductor remain constant.

Mathematical Representation

According to Ohm's law,

$$I \propto V$$

$$I \propto (\text{constant})V$$

$$I \propto KV$$

Where K = constant and it is called conductance of material of conductor.

$$V = \frac{I}{K}$$

$$V = I \times \frac{1}{K}$$

[But $1/K = R$ (resistance of conductor)]

$$V = IR$$

This Law is only valid for metallic conductor at a particular temperature and pressure such conductor are known as ohmic conductors.

Ohm's law is not obeyed by conductors such as Gas discharge tube, Diodes and Thermistors etc.

Resistors and Its Color Codes:

A resistor is a passive two-terminal electronic component designed to introduce a specific amount of resistance into an electrical circuit. It is commonly used to control current or divide voltage in circuit.

In order to find the value of the resistor, generally color coding technique is used where numerical value is represented by a color and the coding is governed by some rules that are given below:

- Color Bands for Digits (1st, 2nd, and 3rd Bands):
- The first and second color bands represent the significant digits of the resistance value.
- The third color band represents the multiplier, indicating the number of zeros to follow the significant digits.

Color Code for Digits:

color	Digit
Black	0
Brown	1
Red	2
Orange	3
Yellow	4
Green	5
Blue	6
Violet	7
Gray	8
White	9

- Multiplier (3rd Band):**

The third band, representing the multiplier, determines the number of zeros to add after the significant digits. For example, if the third band is red, it indicates a multiplier of 10^2 or 100.

- Tolerance (4th Band - Optional):**

The fourth band, if present, represents the tolerance or permissible variation in the resistor's value. It is often a gold or silver band.

Color	Tolerance
None	$\pm 20\%$
Silver	$\pm 10\%$
Gold	$\pm 5\%$
Red	$\pm 2\%$ (rare)
Brown	$\pm 1\%$ (rare)
Green	$\pm 0.5\%$ (rare)

- Temperature Coefficient (5th Band - Optional):**

In some high-precision resistors, a fifth band represents the temperature coefficient, indicating how resistance changes with temperature.

Temperature Coefficient Color Code:

Color	Temperature Coefficient
Brown	100 ppm/ $^{\circ}\text{C}$ ($\pm 0.01\%$)
Red	50 ppm/ $^{\circ}\text{C}$ ($\pm 0.005\%$)
Orange	15 ppm/ $^{\circ}\text{C}$ ($\pm 0.0015\%$)
Yellow	25 ppm/ $^{\circ}\text{C}$ ($\pm 0.0025\%$)
Blue	10 ppm/ $^{\circ}\text{C}$ ($\pm 0.001\%$)
Violet	5 ppm/ $^{\circ}\text{C}$ ($\pm 0.0005\%$)

- To read the resistance value of a resistor using the color bands, follow these steps:
- Identify the first two bands (significant digits).
- Determine the third band (multiplier).
- Calculate the resistance value by combining the significant digits and multiplying them by the multiplier.
- If present, consider the fourth band for tolerance and, if applicable, the fifth band for temperature coefficient.
- For more than four bands the distribution is as follows.

0 1 2 3 4 5 6 7 8 9 0 Black 1 Brown 2 Red 3 Orange 4 Yellow 5 Green 6 Blue 7 Purple 8 Grey 9 White ±1% Brown ±2% Red ±5% Gold ±10% Silver	±1% ±2% ±5% ±10% EXAMPLE: 27K 0 × 1 1 1 × 10 2 2 × 100 3 3 × 1000 4 4 × 10000 5 5 × 100000 6 6 × 1000000 7 7 × 10000000 8 8 × 100000000 9 9 × 1000000000 ±10 ±100	±1% ±2% ±5% ±10% EXAMPLE: 15K 0 0 × 1 1 1 1 × 10 2 2 2 × 100 3 3 3 × 1000 4 4 4 × 10000 5 5 5 × 100000 6 6 6 × 1000000 7 7 7 × 10000000 8 8 8 × 100000000 9 9 9 × 1000000000 ±10 ±100	Temperature Coefficient ±1% 100 50 ±2% 10 5 ±5% 1 5 ±10% 0.5 0.5 EXAMPLE: 620K 0 0 × 1 1 1 1 × 10 2 2 2 × 100 3 3 3 × 1000 4 4 4 × 10000 5 5 5 × 100000 6 6 6 × 1000000 7 7 7 × 10000000 8 8 8 × 100000000 9 9 9 × 1000000000 ±10 ±100
Color Codes	4 Band Resistors	5 Band Resistors	6 Band Resistors

Chapter 10

Resistivity

RESISTIVITY:

The resistance of a material or conductor having unit cubic meter of volume.

OR

It is the resistance of a conductor of unit length and unit area.

OR

Resistivity of a conductor is the resistance of 1 meter long conductor whose area of cross section is 1 meter.

Resistance of a conductor is directly proportional to the length of conductor.

$$R \propto L$$

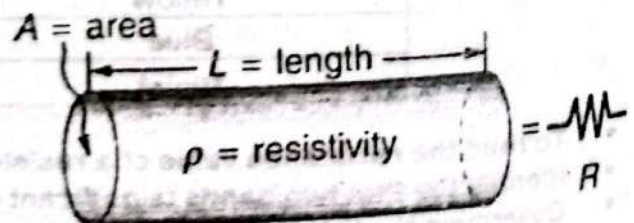
Resistance of a conductor is inversely proportional to area of cross section of conductor.

$$R \propto \frac{1}{A}$$

Combining both equations we get:

$$R \propto \frac{L}{A}$$

$$R = \rho \frac{L}{A}$$



where

ρ = Resistivity of material of conductor

$$\rho = \frac{RA}{L}$$

EFFECT OF TEMPERATURE UPON RESISTANCE:

With the increase in temperature, vibrational motion of the atoms of conductor increases. Due to increase in vibration, probability of collision between atoms and electrons increases. As a result, resistance of conductor increases.

As we know increase in resistance of a conductor is directly proportional to original resistance.

$$\Delta R \propto R_1$$

Change in resistance is directly proportional to change in temperature

$$\Delta R \propto \Delta T$$

Combining (a) and (b)

$$\Delta R \propto R_1 \Delta T$$

$$\Delta R = (\text{constant}) R_1 \Delta T$$

$$\Delta R = \alpha R_1 \Delta T$$

Where α = temperature coefficient

As $\Delta R = R_2 - R_1$ and $\Delta T = T_2 - T_1$

We get

$$\begin{aligned} R_2 - R_1 &= R_1 \alpha (T_2 - T_1) \\ R_2 &= R_1 + \alpha R_1 (T_2 - T_1) \\ R_2 &= R_1 \{1 + \alpha (T_2 - T_1)\} \end{aligned}$$

When $T_1 = 0$ and $T_2 = t$

$$R_t = R_0 [1 + \alpha(t - 0)]$$

$$R_t = R_0 (1 + \alpha t)$$

Resistivity is directly proportional to resistance of material so in term of resistivity above formula can be written as

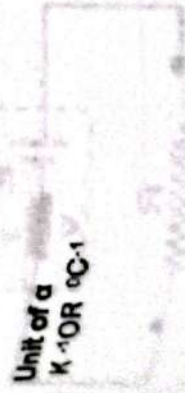
$$\rho_t = \rho_0 (1 + \alpha t)$$

Definition of Temperature Coefficient (α)

Fractional change in resistance per unit original resistance per degree change of temperature is called temperature coefficient.

$$\begin{aligned} \Delta R &= R_0 \alpha \Delta T \\ \alpha &= \frac{\Delta R \Delta T}{R_0} \end{aligned}$$

Unit of α
K⁻¹ OR °C⁻¹



CONDUCTANCE AND CONDUCTIVITY:

	Conductance	Conductivity
Definition	The ease with which a component allows electric current to flow through it.	The ease with which a material allows electric current to flow through it.
Symbol	G	σ
Unit	Siemens (S)	Siemens per meter (S/m)
Dependence	Dimensions of the conductor (length and area), material of the conductor.	Material of the conductor.

**Chapter
10**
Electromotive Force
Internal Resistance:

Internal resistance refers to the opposition to the flow of current offered by the cells and batteries themselves resulting in the generation of heat.

ELECTROMOTIVE FORCE:

It is defined as the potential difference between the terminals of a source, when it is not supplying current to an external electrical circuit.

OR

The EMF of a cell can be defined as the energy converted into electrical energy from other forms (e.g. chemical, mechanical) when unit charge passes through it.

Symbol

It is denoted by "E"

Unit

Its unit is J/coulomb i.e. Volt.

TERMINAL POTENTIAL DIFFERENCE:

It is defined as the potential difference between the terminals of a source, when it is supplying current to an external electrical circuit.

Symbol

It is denoted by "V".

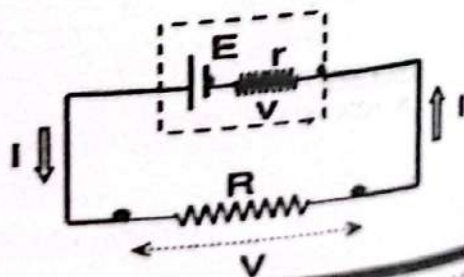
Unit

Its unit is J/coulomb i.e. Volt.

RELATIONSHIP BETWEEN EMF AND TERMINAL POTENTIAL DIFFERENCE:

Consider a source of EMF connected to a resistance "R" through which a steady current "I" flows as shown. When a battery sends the current in external circuit, some amount of current also flows internally through battery.

The internal current comes across some internal resistance (r) by the electrolyte. Thus battery has to use some of its EMF to overcome internal resistance.



Let the

EMF of source = E

Current in circuit = I

Internal resistance = r

External resistance = R

Total resistance = $(R + r)$

Hence

$$\text{Current} = \frac{\text{EMF}}{\text{RESISTANCE}}$$

$$I = \frac{E}{(R + r)}$$

$$E = IR + Ir$$

$$E = V + Ir$$

Where V = terminal potential difference

$$V = E - Ir$$

If Current is Supplied From the Source of EMF (E)

In this condition EMF will be greater than terminal potential difference ($E > V$)

$$V = E - Ir$$

If current is supplied to the source of EMF (E) i.e. Source is being charged

In this condition terminal potential difference will be greater than EMF ($V > E$)

$$V = E + Ir$$

When no current is drawn from the circuit

In this condition $I = 0$ then

$$V = E$$

EFFECT OF INTERNAL RESISTANCE ON CIRCUITS:

Internal resistance in electrical circuits can have several effects depending on the type of circuit and the components involved. Internal resistance primarily arises in voltage sources like batteries and can also be present in other components like wires and connectors. Here are some key effects of internal resistance on circuits:

- Voltage Drop.
- Reduced Output.
- Power Dissipation.
- Voltage Dip under Load.
- Time-Dependent Effects.
- Efficiency Loss.
- Voltage Regulation Challenges.

Sources of EMF:

The primary sources of electromotive force (EMF) are those that directly produce electrical energy. These sources are typically used to provide power to electrical circuits and devices. The primary sources of EMF include:

Chemical Reaction	Chemical Cells, Batteries, Fuel Cells
Light	Solar Cells (Photovoltaic Cells):
Heat	Thermoelectric Generators, Radiolotope Thermoelectric Generators (RTGs), Thermionic Converters, Thermo Couple
Mechanical Energy	Piezoelectric Devices
Mechanical/Magnetic Energy	Generator

Power Dissipation In Resistors:

When an electric current passes through a conductor, some useful electrical energy is dissipated in the form of heat energy. This loss of electrical energy is due to the collision of charges with the atoms of conductor. Loss of electrical energy in unit time is referred to as "power dissipation in resistor"

EXPRESSION FOR POWER LOSS:

Let 'q' amount of electric charge passes through a conductor in unit time, when potential volt "V" is maintaining across its ends.

Hence,

$$\text{Work done by source} = V \times q$$

$$\text{Power} = \frac{\text{Work}}{\text{Time}}$$

$$P = \frac{Vq}{t}$$

$$P = VI$$

$$\therefore I = \frac{q}{t}$$

$$P = I^2 R$$

$$\therefore V = IR$$

$$P = \frac{V^2}{R}$$

$$\therefore I = V/R$$

Unit Of Power

In SI system unit of power is Watt.

Other large units are

- Kilowatt KW (1000 watt or 10^3 watt)
- Megawatt MW (10^6 watt)

Energy Supplied To Resistance:

Heat energy is supplied to a resistance by electric current.

$$\text{Heat energy} = \text{Power} \times \text{Time}$$

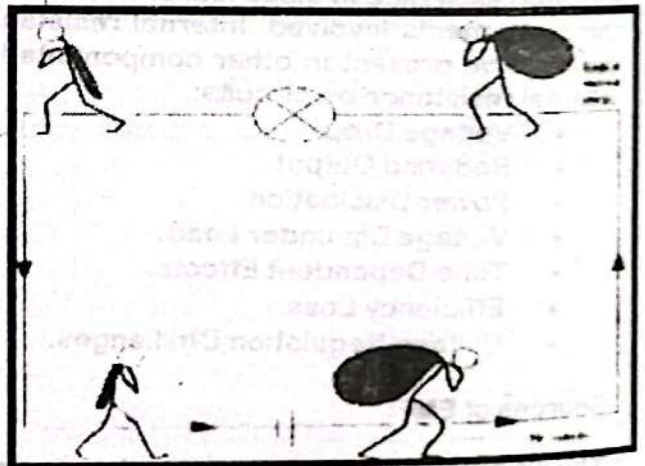
$$\text{Heat energy} = VIt$$

Or

$$\text{Heat energy} = I^2 Rt$$

Or

$$\text{Heat energy} = \frac{V^2 t}{R}$$



PRACTICAL UNIT OF ELECTRICAL ENERGY

Practical unit of heat energy is Kilowatt-Hour (Kwh)

$$1 \text{ Kwh} = 1000 \text{ watt} \times 1 \text{ Hour}$$

$$1 \text{ Kwh} = 1000 \text{ J/Sec} \times 3600 \text{ sec}$$

$$1 \text{ Kwh} = 3.6 \times 10^6 \text{ J}$$

Condition for Maximum Power Transfer:

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The condition for maximum power transfer in an electrical circuit occurs when the load resistance (R_L) is equal to the internal resistance (r) of the voltage source. This principle is known as the Maximum Power Transfer Theorem and is based on the concept of impedance matching. Mathematically, the condition for maximum power transfer is expressed as:

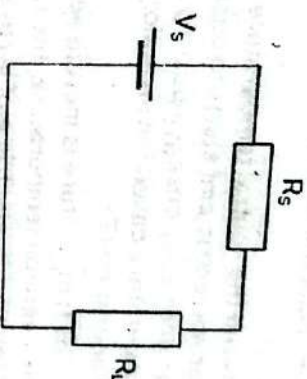
$$R_L = r$$

Consider a circuit in which we have a load resistance R_L and Internal Resistance R_s , then condition for Maximum Power transfer should be:

$$R_L = R_s$$

Using Ohm's Law:

$$I = \frac{V_s}{R_s + R_L} \text{ and } P = I^2 R_L$$



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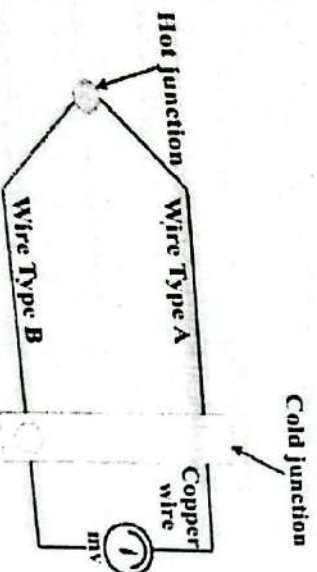
Thermoelectricity

Thermoelectricity:

Thermoelectricity is the conversion of heat energy into electrical energy using a thermocouple.

Thermocouple and its function

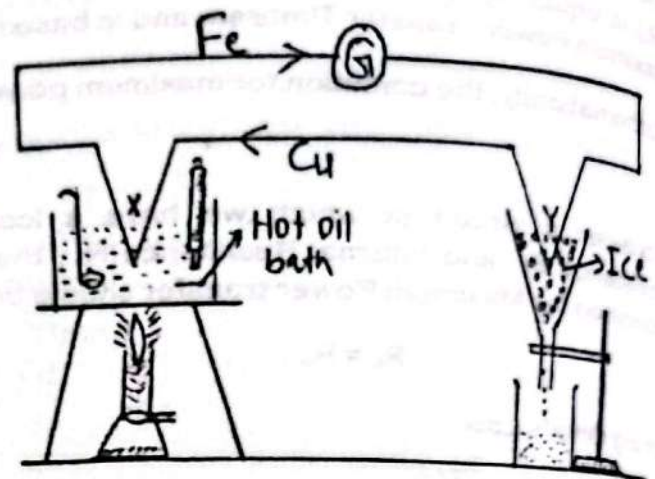
A thermocouple is a type of temperature sensor that converts heat energy into electrical energy. It is made up of two wires made of different metals, which are joined together at one end. When the junction of the wires is heated, a voltage is produced between the free ends of the wires. The amount of voltage produced depends on the temperature difference between the junction and the free ends of the wires.



Variation of thermocouple e.m.f with temperature:
Thermoelectric emf is the voltage generated between two dissimilar metals when their junctions are at different temperatures. This phenomenon is called the thermoelectric effect. The variation of thermoelectric emf with temperature can be studied using an iron-copper thermocouple. One junction is dipped in an oil bath and the other junction is kept at melting ice.

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- When the temperature of both junctions is the same, the thermoelectric emf is zero. This is called cold temperature (T_c).
- As the temperature of the hot junction is increased, the thermoelectric emf increases linearly with temperature.
- At a certain temperature called the neutral temperature (T_n), the thermoelectric emf reaches its maximum value.
- Beyond the neutral temperature, the thermoelectric emf starts to decrease and eventually changes polarity at a temperature called the inversion temperature (T_i).
- As the temperature is increased beyond the inversion temperature, the direction of the thermoelectric emf is reversed.



The inversion temperature depends on the temperature of the cold junction and the types of metals used in the thermocouple.

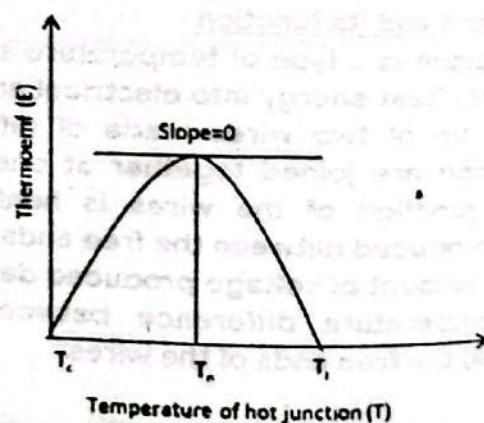
The variation of thermal emf is given by

$$E = \alpha T + \frac{1}{2} \beta T^2$$

Where α and β are constants, whose value depends upon the material of conductor and temperature difference between both junctions.

Graphical Representation: →

Figure: Variation of thermoemf with temperature of hot junction



Relationship between T_i and T_n :

If T_c is the temperature of cold junction, then

Or

$$T_n - T_c = T_i - T_n$$

$$2T_n = T_i + T_c$$

$$T_n = \frac{T_i + T_c}{2}$$

Thermistor:

Thermistor is a resistance whose resistance varies with temperature nonlinearly. It can be used in temperature measurement and control applications. There are two types of thermistors.

1. NTC or Negative Temperature Coefficient: Its resistance decreases with temperature rise
2. PTC or Positive Temperature Coefficient: It increases its resistance if temperature increases.

Application of thermistor in fire alarm

- Thermistor is a variable resistor whose resistance changes with temperature.
- The resistance of the thermistor decreases with increase in temperature.
- In the fire alarm circuit, the thermistor is connected in series with a variable resistor and a transistor.
- The transistor is connected in such a way that it is turned ON when the resistance of the thermistor decreases.
- When the temperature rises above a certain threshold, the resistance of the thermistor decreases, turning ON the transistor.
- This allows current to flow through the buzzer, which produces a beep sound.
- The diode is used to prevent reverse current flow through the transistor.
- The capacitor is used to filter out any noise from the circuit.

Thermostat:

- Thermostat are made up of two Greek words: thermos (heat) and statos (stationary or fixed).
- Thermostats can be used to turn on/off air conditioners, room heaters, and other devices.
- The thermostat circuit consists of a voltage divider circuit and an output "on and off" circuit.
- The voltage divider circuit comprises of a thermistor and a variable resistor.
- The voltage divider circuit output is connected to the base of an NPN transistor through a $1k\Omega$ resistor.
- The voltage divider circuit makes it possible to sense the variation in voltage caused by variation in resistance of thermistor.
- The LED will be switched ON, only if temperature crosses a particular value.

Chapter 10

Kirchhoff's Law

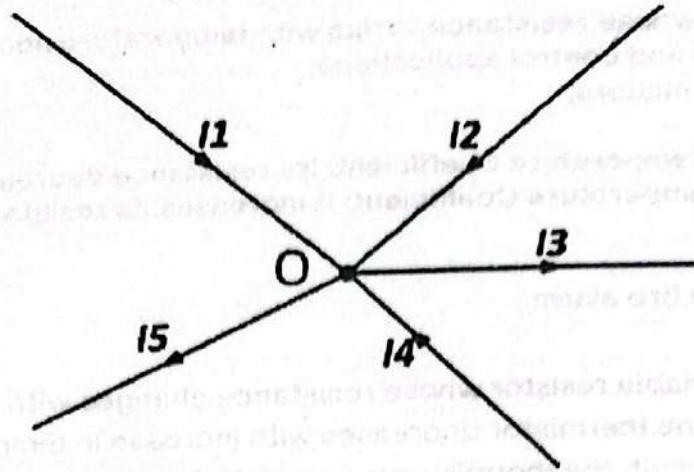
KIRCHHOFF'S CURRENT LAW**Statement:**

In an electrical circuit the algebraic sum of current meeting at any junction is always 0.

Mathematically,

$$\sum I = 0$$

Explanation: Consider a junction O in a circuit as shown in figure there are 3 currents I_1 , I_2 and I_4 entering towards the junction while I_3 and I_5 are leaving from the junction. If entering current is taken positive and leaving current is taken negative, then.



By looking at the above figure we can say that;

$$\sum I = I_1 + I_2 - I_3 + I_4 - I_5 = 0$$

Or

$$I_1 + I_2 + I_4 = I_3 + I_5$$

$$\sum I_{\text{entering}} = \sum I_{\text{leaving}}$$

This law can also be stated as sum of entering current towards a junction is exactly equal to the sum of leaving current.

KIRCHHOFF'S VOLTAGE LAW:

Statement:

The sum of electromotive forces in a loop is equal to the sum of all voltage drops in the loop.

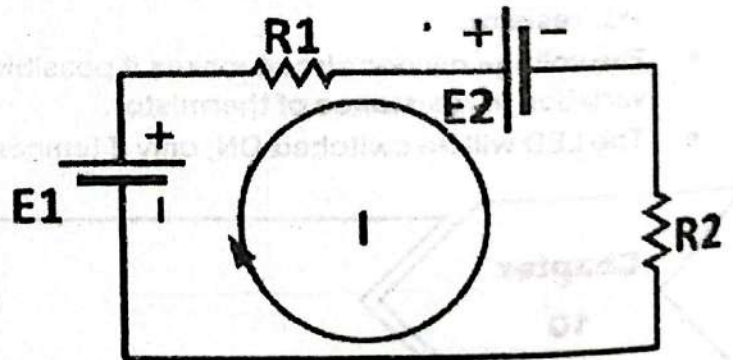
Mathematically,

$$\sum E = \sum V$$

Where $\sum E$ is sum of all e.m.f sources while $\sum V$ represents sum of all voltage drops.

Explanation:

Consider a circuit containing two emf sources (E_1 and E_2) and two resistors (R_1 and R_2). If we start from positive terminal of E_1 and move in clock direction, then both voltage drops (IR_1 and IR_2) will be taken positive, emf source E_1 will be taken positive as direction of current is negative to positive terminal of emf source and it will raise the potential while E_2 will be taken as negative because it will decrease the potential. So, according to KVL



$$E_1 - E_2 = IR_1 + IR_2$$

$$E_1 - IR_1 - E_2 - IR_2 = 0$$

For Single case of battery:

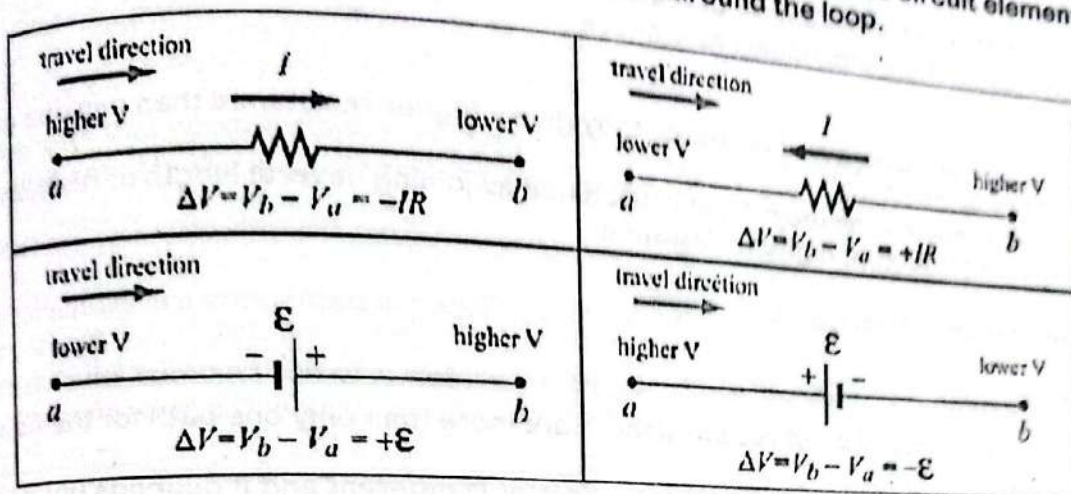
$$E = IR_1 + IR_2$$

e.m.f of battery = sum of potential difference (p.d) across the resistors

$$\sum \Delta V = 0$$

Sign and Direction:

The following conventions apply for determining the sign of delta ΔV across circuit elements. The travel direction is the direction that we choose to proceed around the loop.



Chapter 10

Series and Parallel Combination of Resistance

SERIES AND PARALLEL COMBINATION OF RESISTANCE:

Series Combination Of Resistance:

Characteristics

- 1) In series combination of resistors there is only one path for the flow of electric current.
- 2) Electric current passing through each resistor is same.
- 3) Potential difference across each resistor is different and it depends upon the value or resistance.

Disadvantage Of Series Combination

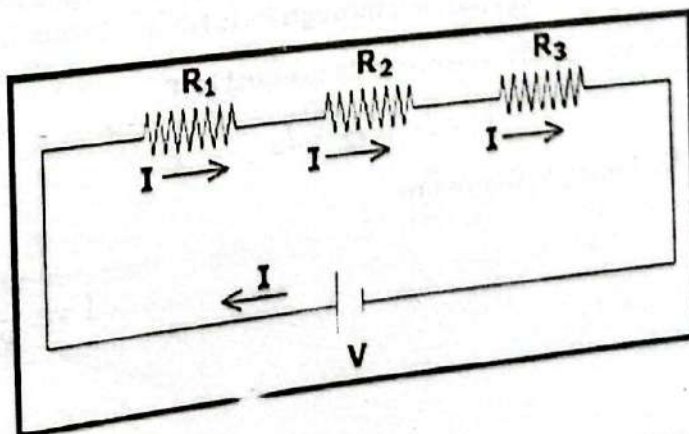
In series combination if one resistor is damaged than other resistors will not work.

Equivalent Resistance Of Circuit

Consider three resistances R_1 , R_2 and R_3 connected to one another in series circuit as shown below

Let the circuit is connected to a power supply of voltage 'V' and an electric current 'I' is passing through the circuit.

Potential difference across R_1 is V_1
 Potential difference across R_2 is V_2
 Potential difference across R_3 is V_3
 The sum of these Potential differences is equal to 'V'.



$$V = V_1 + V_2 + V_3$$

According to Ohm's law $V = IR$
Putting the value of V , we get,

$$IR_{eq} = IR_1 + IR_2 + IR_3$$

Or

$$R_{eq} = R_1 + R_2 + R_3$$

Law: This shows that, when a number of resistances are joined in series their equivalent resistance is equal to the sum of their individual resistances.

Conclusion

1. If resistors are connected in series, they give a higher resistance than any one of the resistors by itself.
2. In series combination their effect is the same as joining several length of resistance wire to form a longer length, higher resistance.

Parallel Combination of Resistance:

Characteristics

- 1) In parallel combination of resistors there are more than only one path for the flow of electric current.
- 2) Electric current passing through each resistor is different and it depends upon the value of resistance.
- 3) Potential difference across each resistor is the same.

Advantage of Parallel Combination

In parallel combination, if one resistance is damaged then the other resistors will work properly because there is more than one path for the flow of electric current.

Equivalent Resistance Of Circuit

Consider three resistances R_1 , R_2 and R_3 connected to one another in parallel circuit as shown below.

Let the circuit is connected to a power supply of voltage ' V ' and an electric current ' I ' is passing through the circuit.

Electric current passing through R_1 is I_1

Electric current passing through R_2 is I_2

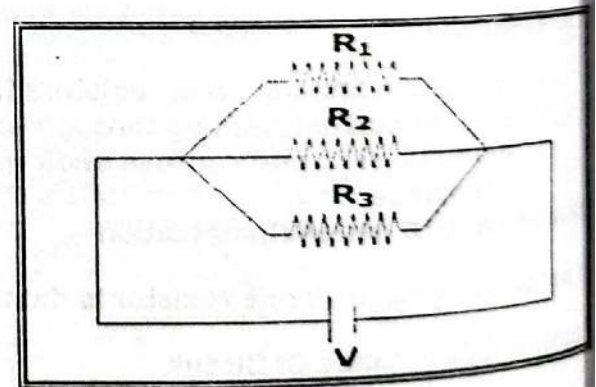
Electric current passing through R_3 is I_3

The sum of all three currents is equal to ' I '

$$I = I_1 + I_2 + I_3$$

According to Ohm's law

$$I = \frac{V}{R}$$



Putting the value of I , we get,

$$\frac{V}{R_{eq}} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

or

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Law: This shows that when a number of resistances are joined in parallel the reciprocal of equivalent resistances is equal to the sum of reciprocal of individual resistances.

Conclusion

- 1) If resistors are connected in parallel, they give a lower resistance than any of the resistors by it self.
- 2) In parallel combination of resistors the effect is the same as using a thick piece of resistance wire instead of a thin wire, a wider conducting path (Large cross sectional area).

Chapter

10

Balance Potential

BALANCE POTENTIAL:

Balance potential is the potential difference between two points in a circuit when there is no current flowing between them. Wheatstone bridge is one of application of balance potential. This is also known as the zero-voltage potential or the quiescent potential

APPLICATION OF THE CONCEPT OF BALANCED POTENTIAL

Some of the Application of the concept of balanced potential are as follows

WHEATSTONE BRIDGE:

It is an instrument proposed by Sir Charles Wheatstone that is used to find the value of unknown resistance accurately.

Structure :

The circuit consists of a diamond-shaped arrangement with four resistances:

1. Two known and fixed resistances, P and Q.
2. One variable but known resistance, S.
3. One unknown resistance, R.

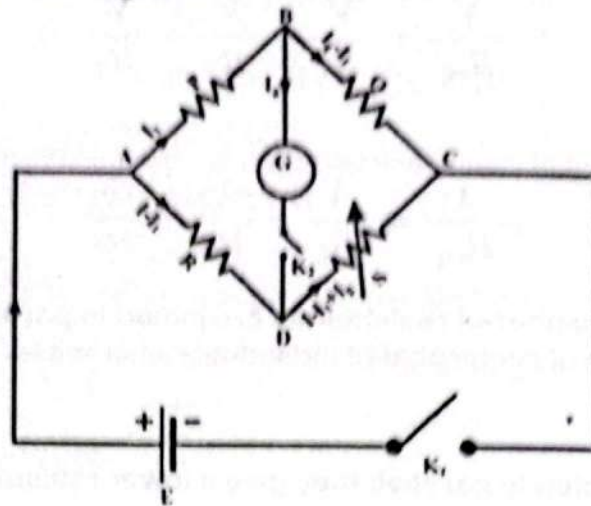
A battery is connected between junctions A and C, while a galvanometer is connected between junctions B and D.

Working Principle:

The Wheatstone Bridge is a null deflection instrument. It achieves a balanced condition by adjusting the variable resistance R until both ends of the galvanometer are at the same potential, resulting in zero current flow through it. The unknown resistance value can then be calculated.

Working and formula:

A wheatstone bridge is shown in the figure.



1. Start by pressing K_1 to connect the battery (EMF E).
2. Next, press K_2 to connect the galvanometer.
3. If the galvanometer shows current, adjust variable resistor R to balance the bridge.
4. Under balance conditions, no current flows through the galvanometer ($I_g = 0$).
5. Now, let's divide the total source current I into two branches P and R :
6. I is divided into I_1 (resistance P) and $(I - I_1)$ (resistance R).
7. The current through resistance Q is $(I - I_1)$.
8. The current through resistance S is $(I - I_1 + I_g)$.

Application of KVL on loop ABDA implies that.

$$I_1 P + I_g G - (I - I_1) R = 0$$

Similarly for loop BCDB we get

$$(I - I_1) Q - (I - I_1 + I_g) S - I_g G = 0$$

Under balanced condition $I_g = 0$ so,

Above Equation become:

$$I_1 P - (I - I_1) R = 0 \text{ or } \frac{I_1}{I - I_1} = \frac{R}{P}$$

And

$$I_1 Q - (I - I_1) S = 0 \text{ or } \frac{I_1}{I - I_1} = \frac{S}{Q}$$

Combining we get

$$\frac{R}{P} = \frac{S}{Q}$$

Or

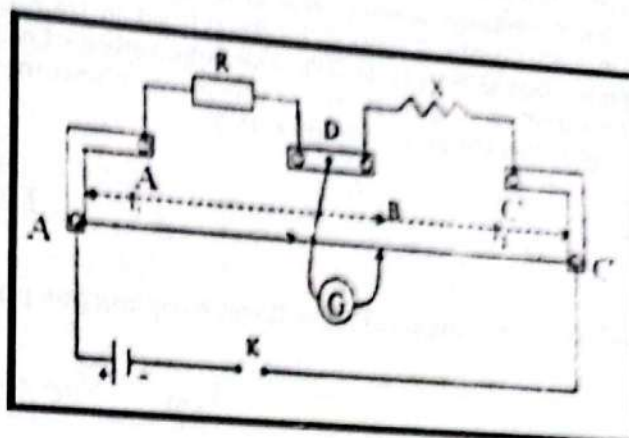
$$\frac{P}{Q} = \frac{R}{S}$$

METER BRIDGE

A meter bridge also called slide-wire bridge is an instrument based on the principle of Wheat stone bridge and it is used to determine unknown resistance.

WORKING AND THEORY

- 1) To find unknown resistance, the key K is closed and a suitable resistance R is taken out from the resistance box.
- 2) The jockey is moved along the wire till for a certain position of galvanometer shows zero deflection, let the point be B.
- 3) Then the length AB is called L_1 and remaining length BC is L_2
- 4) Now if R represents R_1 , X represents R_2 , then L_1 will represent R_3 and L_2 will represent R_4



As for balanced Wheatstone bridge

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

$$\frac{R}{X} = \frac{L_1}{L_2}$$

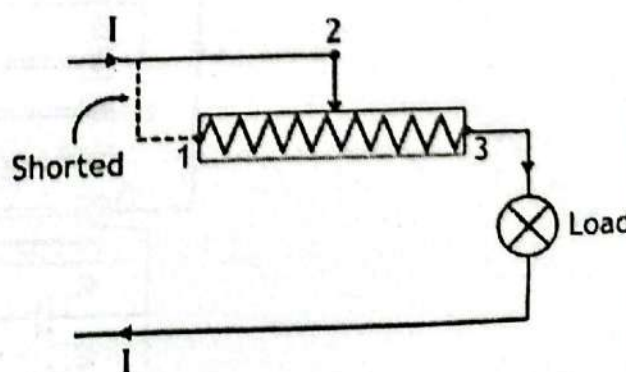
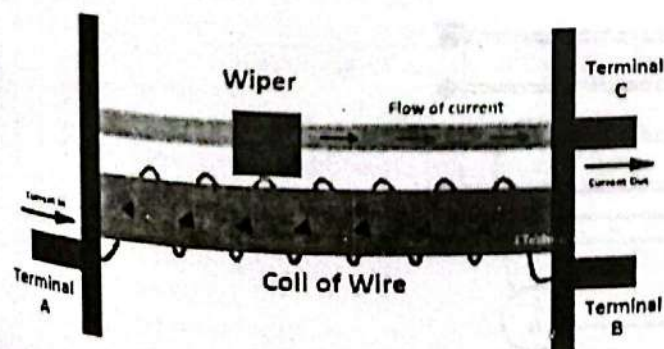
Hence

OR

$$X = \frac{L_2}{L_1} \times R$$

RHEOSTAT:

The rheostat is a wire wound resistance with a slider. End points of wire are connected to terminals A and C while slider is attached with terminal B. Resistance between terminals A and C remains constant while as slider is moved towards C then resistance between A and B increases and resistance between B and C decreases. Rheostat is usually used to control the current.

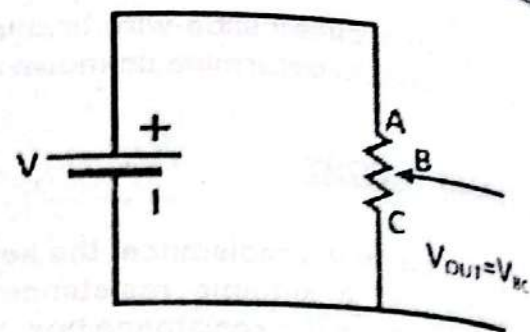


Physics XI

Rheostat as potential divider:

Rheostat can be used as potential divider to provide variable voltage from a fixed voltage source. A voltage source of V volts is connected to the end points A and C while output is being taken across B and C. If R be the resistance of potentiometer from A to C then a constant current I flow through it is given by

$$I = \frac{V}{R}$$



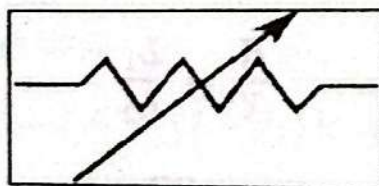
Now output voltage is the voltage drop across portion B to C is given by

$$V_{\text{out}} = V_{BC} = IR_{BC} = \frac{V}{R} R_{BC}$$

As slider is moved towards C then R_{BC} decreases and V_{out} also decreases. Range of V_{out} is from 0 to V .

POTENTIOMETER

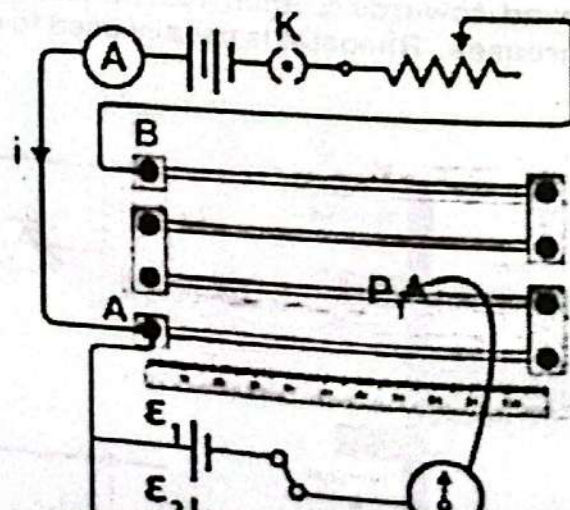
A potentiometer is a device used to measure potential difference between two points in a circuit (i.e., as potential divider) or to compare emf of two cells, without drawing any current from circuit.

CIRCUIT SYMBOL**Working Principle:**

Potential difference across any two points of a wire is directly proportional to the length of wire segment provided conductor has uniform cross-sectional area and a constant current is flowing through it.

Working:

A potentiometer of length AB is shown in the figure.



There are two cells of emf's E_1 and E_2 . The positive ends of the cells are connected to point 'A' and the negative ends of the cells are connected to the jockey through galvanometer (G). To find the null point Key is closed and jockey is adjusted along wire AB where galvanometer (G) doesn't indicate any deflection. Let P_1 be the null point when cell E_1 is connected and corresponding length between the end A of the wire to the null point P_1 be l_1 . The potential difference across this length balances E_1 .

$$E_1 = Kl_1 \dots\dots\dots (i)$$

Where, K is the potential gradient of the wire. Then disconnect the cell of e.m.f E_1 and connect the cell of e.m.f E_2 in the circuit. Let P_2 be the null point and let l_2 be the length between the end A of the wire to the null point P_2 . Then we have

$$E_2 = Kl_2 \dots\dots\dots (ii)$$

By dividing equation (i) by equation (ii), we get

$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$

Hence, by measuring the length l_1 and l_2 we can find the ratio between emf's of two sources. If E_1 is standard emf source, then we can find the emf of other source using following formula.

$$E_2 = \frac{l_2}{l_1} \times E_1$$

Uses of Potentiometer:

1. Compare the emf's of two cells.
2. Measure emf of any cell.
3. Measure internal resistance of a cell.
4. Measure potential differences across a resistor.

Chapter

10

Multiple Choice Questions

MCQ'S		ANS.
1. Kirchhoff's laws are useful in determining: a) current flowing in a circuit b) e.m.f and voltage drops in it a circuit c) power in a circuit d) only e.m.f in a circuit		b
2. The resistance of a superconductor is: a) finite b) infinite c) changes with every conductor d) zero		d
3. Reciprocal of resistance is called: a) conductance b) resistivity c) resonance d) capacitance		a
4. The graphical representation of Ohms law is: a) parabola b) hyperbola c) ellipse d) straight line		d

5. A potential difference is applied across the ends of a wire. If the potential difference is doubled, then the drift velocity of free electrons will:	b) be doubled d) remain unchanged	b
6. Internal resistance is the resistance offered by:	a) Capacitor b) resistor c) Conductor d) source of emf	d
7. Power dissipation in a resistor can be calculated using which formula:	a) $P = V^2 / R$ b) $P = I^2 \times R$ c) $P = V \times I$ d) $P = R / (V \times I)$	b
8. What is a potentiometer primarily used for?	a) Measuring electric current b) Measuring electric charge c) Measuring potential difference (voltage) d) Measuring electric resistance	c
9. A heat-sensitive device whose resistivity changes with the change in temperature is called:	a) conductor b) resistor c) thermistor d) thermometer	c
10. A wire of uniform area of cross-section A length L and resistance R is cut into two parts. The resistivity of each part:	a) Becomes zero b) is halved c) Is doubled d) remains same	d

Exams Practice MCQs

1. According to Ohm's Law, if the voltage across a resistor is doubled, what happens to the current through it?	a) It doubles b) It halves c) It remains the same d) It quadruples	a
2. What is the purpose of color coding on a resistor?	a) To indicate its physical size b) To specify its operating temperature range c) To show its only tolerance value d) To denote its resistance value	d
3. If a resistor has the color bands brown, black, and red, what is its resistance value?	a) 100 ohms b) 1000 ohms c) 10,000 ohms d) 1,000,000 ohms	c
4. What is the SI unit of resistivity?	a) Ohms (Ω) b) Ohm-meters ($\Omega \cdot m$) c) Amperes (A) d) Volts (V)	b
5. If you double the length of a wire while keeping its cross-sectional area constant, how does this affect its resistance?	a) Resistance doubles b) Resistance quadruples c) Resistance remains the same d) Resistance halves	a

6. How does the resistivity of most conductors change with increasing temperature? a) It decreases. b) It remains constant. c) It increases. d) It depends only on the specific material.	c
7. Which unit is used to measure conductance? a) Ohms (Ω) b) Siemens (S) c) Amperes (A) d) Volts (V)	b
8. What is the source of EMF in a battery? a) Chemical reactions b) Magnetic fields c) Friction d) Gravitational force	a
9. In an ideal EMF source, what is the internal resistance? a) Zero b) Infinity c) Variable d) Equal to the external resistance	a
10. In an ideal EMF source with no internal resistance, how does the terminal voltage compare to the EMF when a load is connected? a) Terminal voltage is equal to EMF b) Terminal voltage is greater than EMF c) Terminal voltage is less than EMF d) Terminal voltage is zero	a
11. What is the condition for maximum power transfer from a source to a load resistor in an electrical circuit? a) The source voltage must be maximum b) The load resistance must be minimum c) The load resistance must be equal to the internal resistance of the source d) The load resistance must be maximum.	c
12. What is the Seebeck effect in a thermocouple? a) The generation of an electrical voltage when two different metals are joined and exposed to a temperature difference. b) The change in resistance of a conductor with temperature. c) The thermal expansion of a metal with temperature. d) The production of heat due to electrical current flow.	a
13. What is the mathematical representation of Kirchhoff's Voltage Law (KVL) for a closed loop in an electrical circuit? a) $\sum V = 0$ b) $\sum I = 0$ c) $V = IR$ d) $P = IV$	a
14. When a resistor carries a current (I) of 2 amperes and has a resistance (R) of 4 ohms, what is the power dissipated in the resistor? a) 0.5 watts b) 16 watts c) 4 watts d) 8 watts	d

Physics XI

15. What is thermoelectricity primarily concerned with? a) Conversion of heat into mechanical energy b) Conversion of heat into electrical energy c) Conversion of mechanical energy into heat d) Conversion of electrical energy into heat	b
16. If two resistance of 4 ohm and 8 ohm are connected in series and 2 Ampere current is flowing through 4 ohm resistance what will be current through 8 ohms resistance a) 4 amp b) 8 amp c) 2 amp d) 1 amp	d
17. What is the primary purpose of a Wheatstone bridge circuit? a) To measure voltage b) To measure current c) To measure resistance d) To generate electrical power	c
18. In a potentiometer, what is the principle of measurement used? a) Ohm's Law b) Faraday's Law c) Joule's Law d) Voltage division principle	d
19. Which of the following quantities can be accurately measured using a potentiometer? a) Voltage b) Current c) Resistance d) power	a

QUESTION 5: State the underlying principle of a potentiometer. Describe briefly, giving the necessary circuit diagram, how a potentiometer is used to measure the internal resistance of a given cell?
ANSWER: See lecture (Potentiometer)

Chapter 10

Chapter Formulae & Numericals

Ohm's Law	$V = IR$
Resistivity	$\rho = \frac{RA}{L}$
Dependency of Resistance Upon Temperature	$R_t = R_0(1 + \alpha\Delta T)$
Dependency of Resistivity Upon Temperature	$\rho_t = \rho_0(1 + \alpha\Delta T)$
Conductivity	$\sigma = \frac{1}{\rho}$
Resistance in terms of conductivity	$R = \frac{L}{\sigma A}$
Terminal Potential Difference	$V_t = \varepsilon - Ir$
Power	i) $P = VI$ ii) $P = V^2/R$ iii) $P = I^2R$
Heat Energy Dissipated in a resistance	$E = \text{Power} \times \text{time}$
Maximum Power Transfer in a Circuit	$R_L = r$ $I = \frac{V_s}{R_s + R_L}$

Kirchhoff's First Law

$$P = I^2 R_L$$

Kirchhoff's Second Law

$$\sum I_{in} = \sum I_{out}$$

Wheat Stone Bridge

$$\sum E = \sum V$$

$$\sum \Delta V = 0$$

Potentiometer used to compare E.M.Fs

$$\frac{P}{Q} = \frac{R}{S}$$

Internal Resistance of a cell

$$\frac{E_1}{E_2} = \frac{I_1}{I_2}$$

$$r = \left(\frac{E}{V} - 1 \right) S$$